

HRISHIKESH PINGLE

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Skills

OS Linux, Windows
Languages Java, Python, Dart, TypeScript, C, C++, Rust
Frameworks React/Next.js, Flutter, Svelte, ROS2

Databases MySQL, Firebase, Postgres
Hardware STM32, ESP32, Arduino, RaspberryPI
SCM Git/Github

Experience

Onward Technologies

Jun 2024 – Jul 2024

Software Engineering Intern

Client: Bentley Motors

- Conducted comprehensive Exploratory Data Analysis (EDA) to develop three proof-of-concept (POC) models for the marketing team, resulting in a 40% improvement in targeted campaign effectiveness and engagement metrics.
- Used Jupyter Notebook to collaborate with the team.
- Used Pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-learn, Statsmodels, Plotly for cleaning data, processing data and display the data.

Dashboard

- Engineered an interactive dashboard utilizing Svelte to visualize cloud expenses from Azure, AWS, and GCP APIs, enabling real-time tracking of expenditures while integrating a forecasting feature that accurately predicted monthly costs with an 85% accuracy rate.

WINSpect Technologies Pvt. Ltd.

Dec 2023 – Jan 2024

Systems Software Developer

- Engineered a cross-platform desktop admin suite and Rust-based CLI tool to manage and configure Safran White Rabbit Switch and SecureSync 2400 fleets, enabling centralized control of precision timing infrastructure.
- Built a SvelteKit web frontend providing a unified dashboard for real-time device status, configuration workflows, and fleet-wide monitoring across White Rabbit and PTP-synchronized nodes.
- Designed secure device communication pipelines between the admin interface and Safran hardware, streamlining provisioning and fault diagnostics across distributed timing networks.

Projects

Limit Order Book Simulator - GitHub

Personal

- Engineered a high-performance **limit order book engine in C** implementing price-time priority, scaling to **1M ticks, 25M+ order operations, and 1M+ trades** with **100 heterogeneous agents** (70 noise, 5 market makers, 25 informed) on a single core.
- Achieved **sub-100ns p50 latency** for add/cancel/match operations (match p50 ~ 15ns) and **255K+ ticks/sec** throughput via cache-efficient data structures and profiling-driven optimization.
- Optimized order cancellation from **O(n) → O(1)** using doubly-linked lists and direct node indexing, reducing median cancel latency from microseconds to nanoseconds and enabling deep-book stress tests (**300K+ orders per level**).
- Implemented production-style structures: **red-black tree** for O(log n) price levels, **hash map** for O(1) order lookup, and FIFO queues per level; built-in nanosecond benchmarking with p50/p99 tracking.
- Built a **kdb+ persistence layer in raw C** (KxSystems C API) with async batched I/O (1K orders/flush), agent-tagged orders, and global order IDs; visualized mid-price evolution across **1M ticks** via a Python analytics pipeline.
- The pipeline is stable, does not crash and is deterministically reproducible.

Lock-Free SPSC Order Generator - GitHub

Personal

- Developed a lock-free, cache-aligned **SPSC ring buffer in C++17** using atomic operations and power-of-two indexing.
- Benchmarked at **72M orders/sec** and **sub-100ns p50 queue latency** (p50: 384 cycles) on Intel i7-13620H, with TSC-based measurement and log2 latency histograms.
- Designed for zero consumer stalls and minimal measurement overhead, supporting deep-queue stress tests and percentile latency tracking.
- Provided reproducible build scripts and clear documentation for low-latency, cache-efficient queue design.

PNT Monitoring Tool (Under Development) - Preview

Personal

- Developed a web-based PNT monitoring dashboard inspired by Safran's SecureSync 2400 and White Rabbit, using Svelte, TypeScript, and Tailwind CSS to create a responsive interface for visualizing timing and synchronization data.
- Designed a modular frontend architecture to support future integration of MIB files for real-time display of PNT metrics, such as GNSS-based time synchronization and sub-nanosecond precision data.
- Planned and prototyped alert functionalities to enable proactive monitoring of network timing anomalies, aligning with White Rabbit's high-precision synchronization protocols.

ROS2 Visualization Environment - Workspace - Visualizer

Google Summer of Code 2025

- Developed a modular client-server architecture in ROS 2 (Foxy and Jazzy), visualizing live node communication as Directed Acyclic Graphs (DAGs) using Svelte and Mermaid, with support for dynamic graph updates via service requests.
- Engineered and integrated ROS 2 packages using `rclpy`, `rclnodejs`, C++, and TypeScript; configured build environments with `colcon` and `npm`, and authored complete end-to-end READMEs for reproducibility.
- Proposed advanced visualization enhancements using D3.js and deployment with Tauri, showcasing foresight in scalability and cross-platform support.

Competitions

IMC Prosperity 4 Trading Competition

2026

- Achieved Country Rank 130/1000 teams (Top 13%)
- Achieved Global Rank 750/18000 teams (Top 4%)

Education

Bennett University - 2027

Bachelor of Technology in Computer Science and Engineering

Specialization: Macro and Micro Economics

Certifications

- NVIDIA - MODERN CUDA C++
- Google - Foundations of Cybersecurity
- ISC2 - Security Principles
- IBM - Introduction to Cybersecurity Tools and Attacks
- University of California San Diego - Data Structures
- IBM - Databases and SQL for Data Science with Python